Interface Aesthetics

Ngo, Teo & Byrne · Information Sciences (2003)

Is beauty computable?

Interface aesthetics

Layout quality, written as numbers.

14

Aesthetic measures

Closed-form functions of element bounding boxes

The core claim

Layout quality is not purely subjective.

Balance and spacing become numbers.

Each number is a formula over boxes.

Measured, optimised, checked vs people.

01

The fourteen measures

The measures (1–5)

Balance — even visual weight

Equilibrium — mass near the centre

Symmetry — mirrored about the axes

Sequence — a clear reading flow

Cohesion — aspect ratios agree

Bounding box

Every measure reads position and size.

The measures (6–10)

Unity — parts read as one

Proportion — pleasing ratios

Simplicity — few alignment points

Density — comfortable coverage

Regularity — elements snap to a grid

The measures (11–14)

Economy — few sizes, reused

Homogeneity — evenly distributed

Rhythm — regular, repeating change

Order & complexity — structure wins

regularity — Ngo (2003)

```
# Ngo regularity (alignment term)
regularity = 1 - (n_vap + n_hap) / (2n)

# a tidy 5-bullet column:
# n = 5, n_vap = 1, n_hap = 5
regularity = 1 - (1 + 5) / 10   # 0.40

# slidekit: left edges only
regularity = 1 - 1 / 5          # 0.80
```

Same clean column, two scores

0.40

Ngo x + y regularity

vs

0.80

Left-edge alignment

Why slidekit fits this paper

Every element already has an exact rect.

Measures are arithmetic over geometry.

Deterministic — same deck, same score.

Zero rendering, zero vision.

02

From paper to slidekit

What slidekit adopted

Declined

Regularity

Rhythm

Proportion

Economy

Spacing-regularity

Adopted

Balance

Density

Adoption decisions

Measure	Verdict	Reason
Balance	Adopted	Axis moment
Density	Adopted	Coverage
Regularity	Declined	Halves cols
Proportion	Declined	Ratio noise

A measured-aesthetics lineage

1933	2003	2004	2026
Birkhoff	**Ngo**	**Lok**	**Zhang**
A beauty ratio	Fourteen measures	Visual balance	Moment model

Honest scope

The journal paper is paywalled.

Public copies reproduce only alignment.

Constants are not copied blindly.

No universal beauty equation is claimed.

From measure to code

01 Read geometry

Take each element's rect.

02 Define a formula

Write an open, closed-form score.

03 Verify & ship

Check the source, then report.

"Make design quality auditable."

- Ngo, Teo & Byrne (2003)